Application Architecture Summary

Comprehensive Overview of Service Communication & Amazon EKS Infrastructure

> **Context:** This document synthesizes the multi-layered Amazon EKS architecture diagram, detailing edge routing, authentication, platform orchestration, user workloads, data persistence, security, and CI/CD automation pipelines.

1. Overview & Edge Routing (Layers A – C)

- **Ingress & DNS:** External browser traffic hits Route 53 (`lab.example.com`), passes through AWS WAF for security filtering, and terminates at an Application Load Balancer (ALB) leveraging ACM for TLS certificates.
- **Authorization & Security:** Layer C enforces authentication prior to workload routing via OpenID Connect (OIDC) integration connected with Corporate Identity Providers (IdPs) and enterprise brokers (Keycloak-style OIDC).

2. Platform Services & Orchestration (Layer D)

Core Platform Components

- **JupyterHub & Proxy:** Coordinates multi-user data science environments, authenticating via OIDC and provisioning dynamic, isolated workspace pods for individual users.
- **Argo Workflows / Flyte:** Orchestrates heavy compute workflows, data pipelines, and batch operations.
- **Argo CD:** Implements GitOps-driven deployment, reconciling live cluster states with the version-controlled GitHub repository.
- **Monitoring Service (Prometheus & Grafana):** Scrapes internal metrics and visualizes system observability dashboards.
- **Karpenter:** Handles automated cluster node provisioning and smart horizontal node autoscaling based on live resource demands.

3. Workloads & Data Plane (Layers E – F)

Compute & Storage Architecture

- **User Workloads (Layer E):** Pooled compute workloads, dedicated Jupyter workspace pods, and distributed Ray clusters.
- **Data Persistence (Layer F):**
  - **Amazon RDS:** Manages relational database states (e.g., Hub metadata).
  - **Amazon S3:** Provides scalable object storage and log archives.
  - **Amazon EFS:** Supplies shared network file storage mounted via NFS.
  - **Amazon EBS:** Provides block storage volumes attached via CSI, mounted with Persistent Volume Claims (PVC/RWO).
  - **Amazon ECR:** Stores secure, versioned container images.
- **Private Connectivity:** All data plane interactions route securely via private AWS VPC endpoints (S3, ECR API, DKR, STS).

4. Security (Layer G)

Security & Access Control

- **IAM Roles for Service Accounts (IRSA):** Grants Kubernetes workloads scoped, short-lived AWS permissions without long-lived static credentials.
- **AWS KMS:** Manages encryption keys used to protect data at rest across the platform's storage services.
- **AWS Systems Manager Parameter Store:** Securely stores and retrieves configuration values and secrets used by platform services.

5. Observability & CI/CD Pipelines (Layers H – I)

- **Observability (Layer H):** Prometheus collects metrics, Grafana provides dashboard interfaces, and Grafana Loki aggregates system logs into S3 storage.
- **CI/CD Lifecycle (Layer I):** Code changes committed to GitHub trigger GitHub Actions to build and push container images to Amazon ECR. Argo CD then automatically applies synchronization to the EKS cluster.